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Alkylative Carboxylation of Ynamides and Allenamides with Functionalized Alkylzinc Halides and Carbon Dioxide by a Copper Catalyst

Masanori Takimoto,^{*,[a,b]} Sandeep Suryabhan Gholap,^[a,c] Zhaomin Hou^{*,[a,b,c]}

Abstract: The alkylative carboxylation of ynamides and allenamides with CO₂ and alkylzinc halides catalyzed by a copper catalyst has been developed. A variety of alkylzinc halides bearing functional groups could be used for this transformation to afford α,β -unsaturated carboxylic acids, which contain the α,β -dehydroamino acid skeleton, introducing the corresponding alkyl group and CO₂ across the carbon-carbon triple or double bond. This alkylative carboxylation formally consists of Cu-catalyzed carboxylation of ynamides or allenamides with alkylzinc halides and the subsequent nucleophilic carboxylation of the resulting alkenylzinc species with CO₂. This protocol would be a useful method for the synthesis of α,β -dehydroamino acid derivatives possessing a functionalized alkyl group because the high regio- and stereoselectivity, simple one-pot procedure as well as the use of CO₂ as a starting material.

Introduction

Utilization of carbon dioxide (CO₂) as a C1 building block for the synthesis high value-added chemicals has attracted increasing interest due to its natural abundance, low cost, inherently renewable nature and environmentally benign character. Carboxylic acids are useful synthetic intermediates in organic synthesis and also important structural motifs frequently found in numerous biologically relevant compounds and functional materials.^[1] Therefore, development of efficient methods to utilize CO₂ for preparing synthetically valuable carboxylic acids is highly important. Because of the relatively higher thermodynamic stability of CO₂, conventional carboxylation protocols with CO₂ are conducted by using highly nucleophilic organometallic species, such as organolithium and Grignard reagents.^[2] The use of such highly reactive species limits the scope of those protocols due to poor chemoselectivity. In the

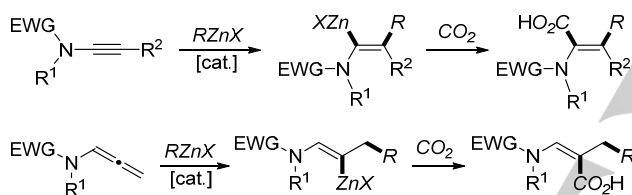
past decades, significant advances in studies on transition metal-catalyzed carboxylation reactions with CO₂ have been achieved, and catalytic carboxylation reactions which can be performed under mild conditions with high chemoselectivity have been reported.^[3] Among various types of catalytic carboxylation methods, the carboxylation reactions of unsaturated compounds such as alkynes and allenes with CO₂ (eg. Hydrocarboxylation,^[4] alkylative/arylate carboxylation,^[5] and heterocarboxylation^[6]) have been extensively studied,^[7] as these reactions can provide unsaturated carboxylic acids or derivatives, which are valuable building blocks in organic synthesis. Moreover, such methods have interesting benefits on constructing molecular skeletons; in principal, the carboxylation of unsaturated compounds bearing a C-C multiple bond can make multiple covalent bonds in a single transformation process. For instance, in the alkylative carboxylation of alkynes, both an alkyl group and CO₂ add to a C-C triple bond with two C-C bond formation to give α,β -unsaturated carboxylic acids bearing a β -branched alkyl group. This straightforward transformation is desirable for the synthesis of highly substituted α,β -unsaturated carboxylic acids in controlled regio- and stereochemistry. However, the scope of the alkyl groups that could be introduced on the carboxylation products in the catalytic reactions previously reported is still limited to relatively simply alkyl groups, such as methyl and ethyl groups. Therefore, exploration of the alkylative carboxylation of unsaturated compounds with CO₂ which can introduce various alkyl groups including functionalized ones is of obvious interest and importance.

α,β -Dehydroamino acids are a class of α,β -unsaturated carboxylic acids bearing a nitrogen atom on their α - or β -carbon. They are also important components found in several biologically active natural products.^[8] Moreover, dehydroamino acids and their analogues can serve as useful precursors in organic synthesis, such as in the preparation of non-natural amino acid derivatives.^[9] Thus, development of the methods to synthesize dehydroamino acids and derivatives by carboxylation reactions using CO₂ is highly important. Our group has recently shown that N-heterocyclic carbene (NHC) copper complexes can serve as excellent catalysts for the carboxylation of various organic substrates with CO₂.^{[5b,d,e,i],[6c],[10]} In this context, we have achieved the alkylative carboxylation of ynamides with dialkylzinc reagents (R₂Zn) and CO₂ which affords α,β -dehydro- α -amino acid esters bearing a branched alkyl group introduced by the addition of dialkylzinc reagents.^[5e] In this reaction, the alkylative carboxylation formally consists of Cu-catalyzed carboxylation of ynamides and the subsequent carboxylation of resulting alkenylzinc species with CO₂. More recently, we developed also the alkylative carboxylation of allenamides with dialkylzinc reagents and CO₂ which affords α,β -dehydro- β -amino

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acid esters through the similar reaction pathway.^[5] Although these methods can afford a variety of α,β -dehydroamino acid esters with well-controlled configurations, the introducible alkyl groups are limited to methyl, ethyl, or butyl due to the accessibility of dialkylzinc reagents. To expand the scope of the reactions, the use of alkylzinc halides ($RZnX$) is preferable because alkylzinc halides can be easily prepared by the reaction of the corresponding alkyl halides with metallic zinc.^[11] Moreover, the less nucleophilic nature and higher functional group tolerability of alkylzinc halides allow coexistence of some functional groups, such as ester, ether, and nitrile groups on their alkyl moiety,^[12] which might enable the introduction of functionalized alkyl groups during the carboxylation process. These features of alkylzinc halides prompted us to develop a new alkylative carboxylation protocol for ynamides and allenamides using alkylzinc halides as the alkylating agents by the combination of carbozincation with alkylzinc halides and the subsequent carboxylation of the resulting alkenylzinc halide species with CO_2 (Scheme 1). Herein, we report our studies on the alkylative carboxylations of ynamides and allenamides^[13-18] with alkylzinc halides and CO_2 , which afford various α,β -dehydro- α - and β -amino acid esters with well-controlled configuration.



Scheme 1. Alkylative carboxylation of ynamides and allenamides based on catalytic carbozincation reaction.

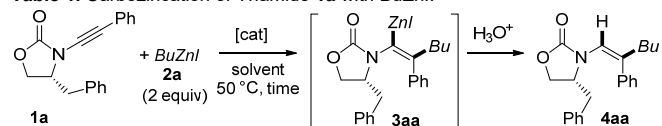
Results and Discussion

Alkylative Carboxylation of Ynamides

As illustrated in Scheme 1, we planned to develop the alkylative carboxylation process based on the carbozincation with alkylzinc halides and the subsequent nucleophilic carboxylation. Our investigation started by studying the butylative carboxylation of ynamide **1a** using $BuZnI$ as the alkylating reagent. We first screened the reaction conditions for the initial carbozincation process (Table 1).^[17] Initially, the reaction of **1a** with $BuZnI$ (**2a**) was tried in THF using $[(IPr)CuCl]$ as the catalyst as we have demonstrated that it was the excellent catalyst for the carbozincation and alkylative carboxylation of ynamides using dialkylzinc reagents (R_2Zn).^[5e] However, the reaction of **1a** with $BuZnI$ (2 equiv) in the presence of $[(IPr)CuCl]$ (5 mol%) in THF was sluggish even at 50 °C; the desired butylated product **4aa** was obtained in 49% yield after 48 hr via hydrolysis of **3aa** along with the recovery of 44% of **1a** (Table 1, entry 1). In contrast, a relatively simple copper salt, $Cu(acac)_2$, showed higher catalytic activity; the reaction of **1a** with $BuZnI$ (2 equiv) in the presence of $Cu(acac)_2$ (10 mol%) under similar reaction conditions (50 °C, 48 hr) gave the butylated product **4aa** in 80% yield (Table 1,

entry 2). We next examined the carboxylation of **3aa** prepared in THF under these conditions (Table 2, entries 1-4). The reaction of **3aa** alone with CO_2 (1 atm) in THF at 50 °C was sluggish to provide the desired carboxylation product **5aa** only in 20% yield after 48 hr (Table 2, entry 1). The addition of a catalytic amount of IPr (10 mol%) or a stoichiometric amount of CsF (1.0 equiv) showed almost no effect on the yield of **5aa** (Table 2, entries 2 and 3).^[19] As Kondo et al. reported that the carboxylation of alkylzinc halides proceeded well in DMF in the presence of a stoichiometric amount of $LiCl$,^[20] we examined the carboxylation process with $LiCl$ and DMF (Table 2, entry 4). As the result, the yield of **5aa** was improved to 60% by the addition of $LiCl$ (2.0 equiv) and DMF (THF/DMF = 1:1) although the reaction still not completed even after 60 hr as 24% of uncarboxylated product **4aa** was obtained. To accelerate the carboxylation process more, the carbozincation and the carboxylation processes were examined in DMF alone (Table 1, entries 3-4 and Table 2, entries 5-8). When DMF was used as the solvent for the carbozincation process, the reaction was slightly deterred and 10 mol% of $Cu(acac)_2$ was not enough to complete the carbozincation process even in 48 hr at 50 °C (Table 1, entry 3). However, increase of the loading of $Cu(acac)_2$ to 35 mol% was effective to accomplish the carbozincation process in 24 hr (Table 1, entry 4). The subsequent carboxylation process also completed smoothly in 48 hr in the presence of $LiCl$ (2 equiv) at 50 °C to give the desired carboxylation product **5aa** in 75% yield as the result of sequential butylative carboxylation reaction (Table 2, entry 5). When the carboxylation was performed in DMF in the absence of $LiCl$, **5aa** was obtained only in 23% yield (Table 2, entry 6). The addition of $LiCl$ prior to the carbozincation process has no adverse effect and both the carbozincation and the carboxylation reactions smoothly proceeded without loss of the yield of **5aa** (Table 2, entry 7). The reaction could be carried out with 25 mol% of $Cu(acac)_2$, and moreover, the carboxylation process could be accomplished in 24 hr to provide **5aa** in 76% yield when the reaction was carried out at 60 °C (Table 2, entry 8).

Table 1. Carbozincation of Ynamide **1a** with $BuZnI$.



Entry	[cat] (mol%)	Solvent	Time [h]	4aa [yield, %]	1a [recovery, %]
1	$[(IPr)CuCl]$ (5)	THF	48	49	44
2	$Cu(acac)_2$ (10)	THF	48	80	-
3	$Cu(acac)_2$ (10)	DMF	48	64	27
4	$Cu(acac)_2$ (35)	DMF	24	81	-

Table 2. Butylative carboxylation of ynamide **1a**

Entry	Cu cat. [mol%]	Solvent	Time A [h]	Additive (equiv)	Time B [h]	5aa [%]	4aa [%]
1	10	THF	48	-	48	20	58
2	10	THF	48	IPr (0.1)	48	26	53
3	10	THF	48	CsF (1.0)	48	31	60
4	10	THF	48	LiCl (2.0)/DMF ^[a]	60	68	24
5	35	DMF	24	LiCl (2.0)	48	75	-
6	35	DMF	24	-	48	23	62
7 ^[b]	35	DMF	24	LiCl (2.0)	48	76	-
8 ^{[b],[c]}	25	DMF	24	LiCl (2.0)	24	78	-

[a] DMF was added after the carboxylation process with LiCl. The resulting ratio of THF to DMF is 1:1. [b] The carboxylation was carried out in the presence of LiCl after mixing of LiCl with **1a**, BuZnI, and Cu(acac)₂ in DMF. [c] The carboxylation process was performed at 60 °C.

With the optimized reaction conditions in hand, the scope of organozinc halides was examined using ynamide **1a** as the substrate (Table 3). When alkylzinc iodide **2b** bearing a hydroxyl group protected by a silyl ether group (TBDMSO, TBDMS = *tert*-butyldimethylsilyl) were used, the corresponding alkylative carboxylation product **5ab** was obtained in 76% yield (Table 3, entry 1). Similarly, the reaction using alkylzinc iodide **2c** bearing a benzoyloxy group also proceeded smoothly to provide the desired product **5ac** in 79% (Table 3, entry 2). The alkylative carboxylation using alkylzinc iodide **2d** bearing an ester group with α -methylene protons gave the alkylative carboxylation product **5ad** in 56% yield under the standard conditions along with 35% of the uncarboxylated product (Table 3, entry 3). When the carboxylation step was repeated at 50 °C for 48 hr, the yield of **5ad** was improved to 77% (Table 3, entry 4). In addition to alkylzinc iodides, also alkylzinc bromides, which can be easily prepared from less expensive alkyl bromides, can be used in this alkylative carboxylation protocol as the alkylating reagents. The reaction of **1a** with alkylzinc bromide **2e** smoothly proceeded under the similar conditions to give the desired product **5ae** in 72% yield (Table 3, entry 5).

Table 3. Alkylative carboxylation of **1a** with various alkylzinc halides

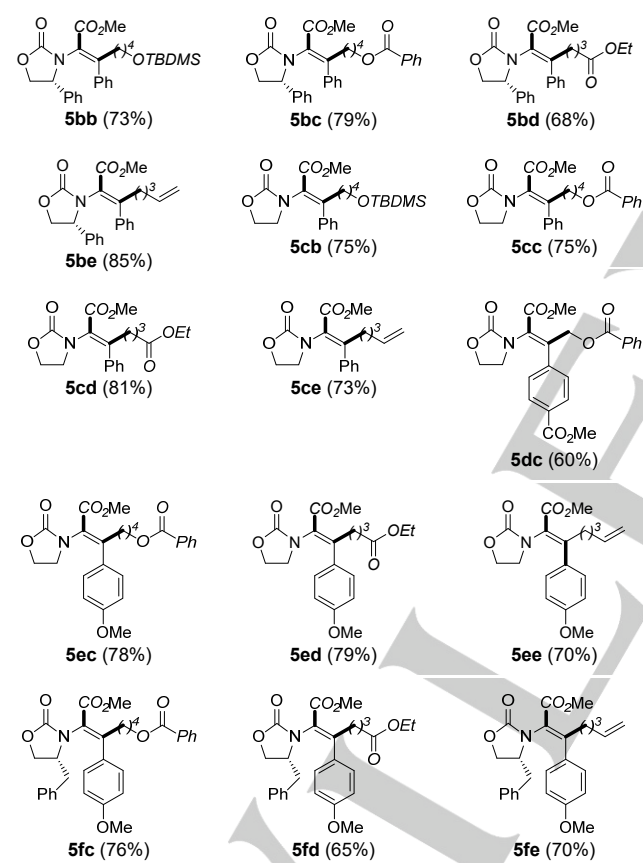
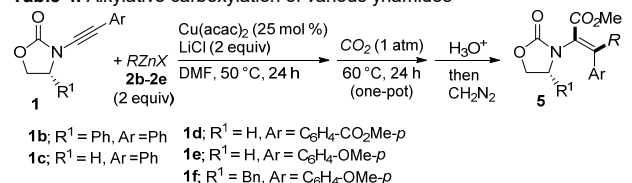
Entry	RZnX	Product (5a)	Yield [%]
1	TBDMSO-(CH ₂) ₄ ZnI 2b		76
2	Ph-C(=O)-O-(CH ₂) ₄ ZnI 2c		79
3	EtO-C(=O)-CH ₂ -CH ₂ -ZnI 2d		56 ^[a]
4 ^[b]	2d	5ad	77
5	CH ₂ =CH-CH ₂ -ZnBr 2e		72

[a] 35% of uncarboxylation product was obtained. [b] The reaction was carried out by using 35 mol% of Cu(acac)₂ and the carboxylation process was performed at 50 °C for 48 hr.

Next, the scope of ynamide substrates was explored (Table 4).^[21] The reaction of ynamide **1b** bearing a phenyl-substituted chiral oxazolidinone moiety with functionalized alkylzinc halides **2b**, **2c**, **2d** and **2e** smoothly proceeded to afford the desired products **5bb** (73%), **5bc** (79%), **5bd** (68%), and **5be** (85%) in good to high yields, respectively. Also the sterically less demanding ynamide **1c** was found to be a good substrate in this alkylative carboxylation protocol. The reactions of **1c** with functionalized alkylzinc halides **2b**, **2c**, **2d**, and **2e** gave the corresponding alkylated carboxylic acid esters **5cb** (75%), **5cc** (75%), **5cd** (81%), and **5ce** (73%) in good yields. Ynamide **1d** bearing a *p*-methoxycarbonylphenyl group on the alkynyl moiety could be tolerated to the present reaction conditions and the reaction of **1b** with alkylzinc iodide **2c** and CO₂ gave **5dc** in 60% yield. Ynamide **1e** bearing a *p*-methoxyphenyl group attached to the alkyne moiety could participate the alkylative carboxylation reaction with alkylzinc halide **2c** to give the desired products **5ec**

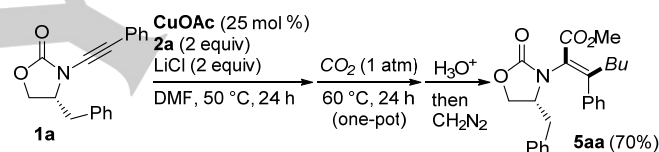
in slightly higher yield (78%) than that of **5dc**, which suggests that the electron-donating group on the phenyl substituents might be favorable for the current reactions. Indeed, the alkylative carboxylation reactions of ynamide **1e** with other alkylzinc halides, such as **2d** and **2e**, proceeded well to give **5ed** and **5ee** in 79% and 70% yields, respectively. Similarly, the reactions of ynamide **1f** with alkylzinc halides **2c**, **2d**, and **2e** afforded **5fc** (76%), **5fd** (65%), and **5fe** (70%) in good yields.

Table 4. Alkylative carboxylation of various ynamides

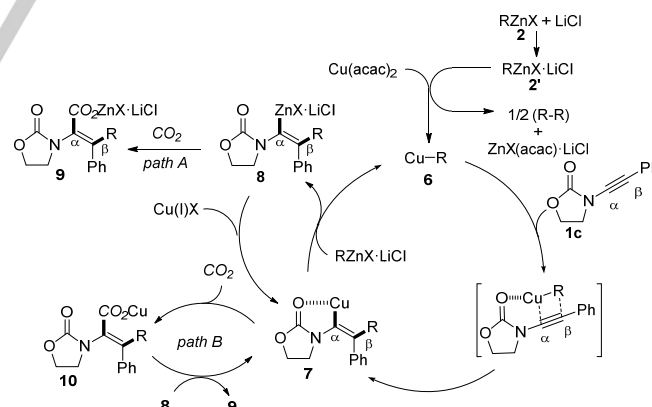


The use of Cu(I) salt, CuOAc, as the catalyst instead of Cu(acac)₂ in the alkylative carboxylation of ynamide **1a** with BuZnI and CO₂ resulted in the formation of **5aa** in 70% yield (Scheme 2), which is comparable with that observed in the reaction using Cu(acac)₂ (Table 2, entry 8). This control experiments suggests that Cu(I) species, which would be formed

by the reduction of Cu(II) with alkylzinc halides, would act as the catalytically active species in the current reaction system. The possible reaction pathway is illustrated in Scheme 3. Initially, Cu(acac)₂ salt would react with alkylzinc halides-LiCl adduct **2'**, which would be formed from alkylzinc halide and LiCl, to give alkylcopper(I) species. The carbonyl group on the ynamide moiety would act as a directing group in the subsequent carbocupration process to form the alkenylcopper species **7** in a regioselective manner attaching the copper unit and alkyl group onto α- and β-carbons, respectively.^{[5e], [16-17]} Transmetalation between **7** and an alkylzinc halide would provide an alkenylzinc species **8** regenerating alkylcopper(I) species. For the subsequent carboxylation process, there would be two possible pathways. According to the previous report by Kondo et al.^[20a] alkenylzinc-LiCl adduct **8** itself would have enough nucleophilicity toward CO₂ under the current reaction conditions. Therefore, the direct carboxylation of **8** with CO₂ would give zinc carboxylate **9**, which could give the corresponding carboxylic acid ester after hydrolysis and esterification (Scheme 3, path A). In another pathway (Scheme 3, path B), transmetalation of alkenylzinc species **8** with Cu(I) species would form alkenylcopper species **7**, in which CO₂ insertion into copper-carbon bond would take place to afford copper carboxylate **10**. Transmetalation of **10** with alkenylzinc species **8** would afford zinc carboxylate **9** regenerating alkenylcopper **7**.



Scheme 2. Alkylative carboxylation using CuOAc as the catalyst



Scheme 3. Possible reaction pathway for the alkylative carboxylation of ynamide **1c**.

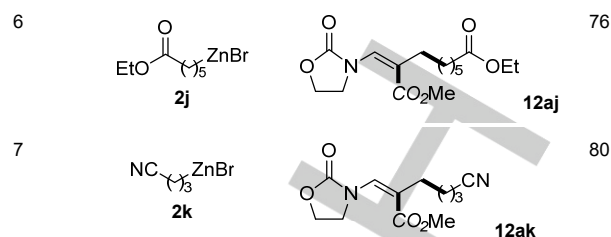
Alkylative Carboxylation of Allenamides

We next studied the alkylative carboxylation of allenamides with alkylzinc halides with CO₂ composed of the carbocation and the subsequent carboxylation processes. In our initial studies on

the butylative carboxylation of **11a** with butylzinc bromide (**2f**), we found that a protocol similar to that developed for the above-mentioned alkylative carboxylation of ynamides worked well also for allenamides (for details, see supporting information). In the presence of Cu(acac)₂ (35 mol%) and LiCl (2.0 equiv), carbocation of **11a** completed after 48 hr at 50 °C in DMF and the subsequent treatment of the resulting reaction mixture with CO₂ (1 atm, at 50 °C, for 48 hr) gave the desired butylative carboxylation product **12af** in 83% yield (Table 5, entry 1). Under these optimized conditions, the alkylative carboxylation of **11a** can be carried out with various organozinc bromides (Table 5, entries 2-7). The reactions with relatively simple alkylzinc bromides, such as **2g**, **2h**, and **2e**, proceeded smoothly to afford the desired products **12ag** (68%), **12ah** (76%), and **12ae** (68%), in good yields (Table 5, entries 2-4). The silyl ether moiety on alkylzinc bromide **2i** (TIPS = triisopropylsilyl) was tolerated under the current reaction conditions (Table 5, entry 5) to give **12ai** in 71% yield. Also functionalized alkylzinc halides **2j** and **2k** bearing an ester group and a nitrile group could participate in the current alkylative carboxylation reaction to give the desired product **12aj** and **12ak** in 76% and 80% yields, respectively (Table 5, entries 6 and 7).

Table 5. Alkylative carboxylation of **11a** with various alkylzinc bromides

Entry	RZnBr (2)	Product (12a)	Yield [%]
1	BuZnBr 2f	 12af	83
2	Cy(CH ₂) ₂ ZnBr 2g	 12ag	68
3	Ph(CH ₂) ₃ ZnBr 2h	 12ah	76
4	CH ₂ =CH(CH ₂) ₃ ZnBr 2e	 12ae	68
5	TIPSO(CH ₂) ₃ ZnBr 2i	 12ai	71

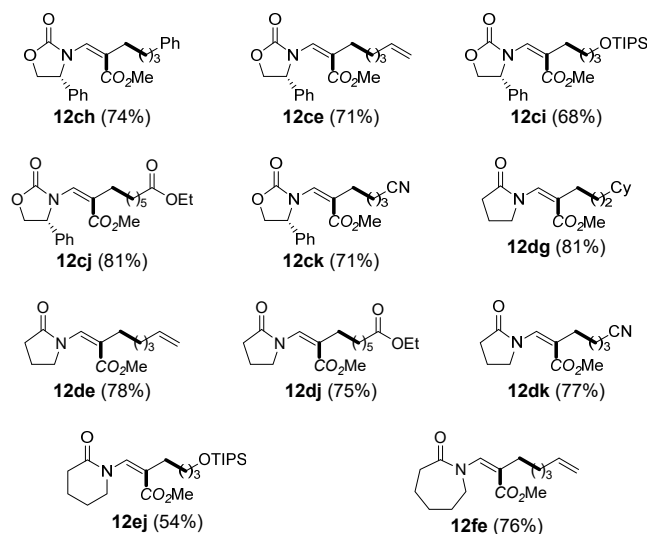


The studies on the scope of allenamide substrates in the present alkylative carboxylation protocol are summarized in Table 6.^[22] Allenamide **11b** bearing an Evans' oxazolidinone moiety underwent the alkylative carboxylation using alkylzinc bromides **2g**, **2e**, **2i**, **2j**, and **2k** under the standard conditions to afford the desired products **12bg** (68%), **12be** (82%), **12bi** (68%), **12bj** (77%), and **12bk** (76%) in good to high yields, respectively. Similarly, allenamide **11c** having a phenyl-substituted chiral oxazolidinone moiety could smoothly react with the various alkylzinc bromides and CO₂ to provide the corresponding alkylative carboxylation products **12cg** (68%), **12ch** (74%), **12ce** (71%), **12ci** (68%), **12cj** (81%) and **12ck** (71%). Alkylative carboxylation reactions of allenamide **11d** possessing a γ-lactam unit with alkylzinc bromides **2g**, **2e**, **2j**, and **2k** could cleanly proceed to give the desired products **12dg** (81%), **12de** (78%), **12dj** (75%), and **12dk** (77%) in high yields despite the presence of relatively acidic protons next to the carbonyl group of its lactam moiety. Allenamides **11e** and **11f** which had a six- and seven-membered lactam moiety were also tolerated under the reaction conditions. The reaction of **11e** with alkylzinc bromide **2i** with CO₂ gave **12ej** in 54% yield and that of **11f** with **2e** and CO₂ afforded **12fe** in 76% yield.

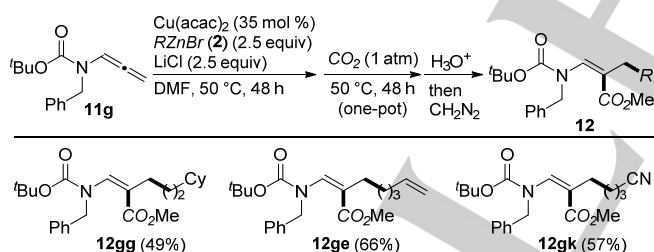
Table 6. Alkylative carboxylation of various allenamides

11	R ¹	12	R ¹
11b ; n = 1, X = O	R ¹ = Ph	11d ; n = 1, X = CH ₂	R ¹ = H
11c ; n = 1, X = O	R ¹ = Bn	11e ; n = 2, X = CH ₂	R ¹ = H
		11f ; n = 3, X = CH ₂	R ¹ = H

 12bg (68%)	 12be (82%)	 12bi (68%)
 12bj (77%)	 12bk (76%)	 12cg (68%)



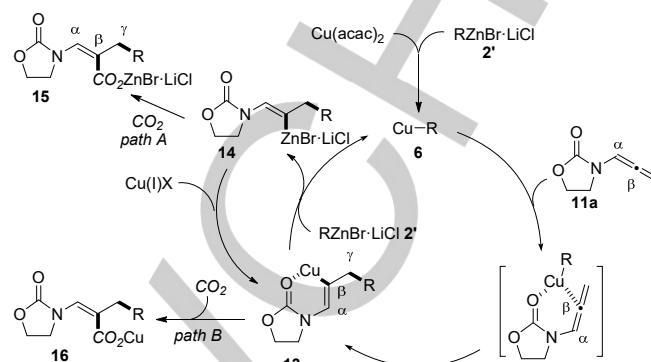
Acyclic BOC-protected allenamide **11g** also underwent the present alkylative carboxylation (Scheme 4). The carboxylation of **11g** with alkylzinc bromide **2g** and the subsequent carboxylation reaction of the resulting alkenylzinc species at 50 °C gave the corresponding alkylative carboxylation product **12gg** in 49% yield. The reaction of **11g** with alkylzinc bromide **2e** bearing a terminal alkenyl group proceeded well in the similar manner to afford the desired product **12ge** in 66% yield. Also functionalized alkylzinc bromide **2k** having a nitrile group could be used in the current sequential transformation to obtain the desired alkylative carboxylation product **12gk** in 57% yield.



Scheme 4. Alkylative carboxylation of acyclic allenamide **11g**.

A possible reaction pathway for the present alkylative carboxylation of allenamide **11a** is shown in Scheme 5. The reaction would proceed through the processes similar with those for the above-mentioned alkylative carboxylation of ynamides except the regioselectivity. In the reaction of alkylcopper(I) species **6** with allenamide **11a**, the amide carbonyl group could work as a directing group to introduce the copper unit and alkyl group onto β - and γ -positions, respectively to give the alkenylcopper species **13**. Transmetalation of **13** with the $RZnBr \cdot LiCl$ **2'** would provide an alkenylzinc species **14** regenerating alkylcopper(I) **6**. The alkenylzinc species **14** might

be able to react with CO_2 directly to afford zinc carboxylate **15** (Scheme 5, path A). Alternatively, CO_2 addition might occur also on alkenylcopper **13** (Scheme 5, path B) in the similar fashion previously discussed in Scheme 3.



Scheme 5. Possible reaction pathway for the alkylative carboxylation of allenamide **11a**.

Conclusion

In summary, we have developed an efficient protocol for the regio- and stereoselective alkylative carboxylation of ynamides and allenamides with various alkylzinc halides and CO_2 in the presence of LiCl by using a copper catalyst. This transformation proceeds through the Cu-catalyzed carboxylation and the subsequent LiCl-promoted carboxylation of the resulting alkenylzinc halide species. This sequential reaction can be carried out easily in one-pot under mild conditions to afford a variety of α,β -dehydroamino acid derivatives in good to high yields. The use of easily available and functional group tolerant organozinc halides enables the introduction of various alkyl groups including those bearing functional groups such as ester, silyl ether, and nitrile groups onto the corresponding carboxylation products. In view of the high regio- and stereoselectivity for the formations of substituted α,β -dehydro- α - and β -amino acid derivatives with well-defined configurations, simple one-pot operation, and the use of CO_2 as a starting material, this protocol may serve as a practically useful and attractive method in synthetic organic chemistry.

Experimental Section

Typical procedure for alkylative carboxylation of ynamides: methyl (2Z)-2-((R)-4-benzyl-2-oxooxazolidin-3-yl)-3-phenylhept-2-enoate (4aa) (Table 2, entry 8): In a Schlenk flask equipped with a PTFE J. Young valve (RA-type) and a magnetic stirring bar, $Cu(acac)_2$ (16.5 mg, 0.063 mmol), ynamide **1a** (50 mg, 0.18 mmol), LiCl (15.3 mg, 0.36 mmol) were added and dissolved in dry DMF (1.8 mL). To this was added a solution of $BuZnI$ (**2a**) (1.0 M in THF, 0.36 mL, 0.36 mmol) at 0 °C. The resulting mixture was stirred at 50 °C for 24 hrs. After the reaction mixture was cooled to 0 °C, the flask was evacuated and CO_2 (stored in balloon) was quickly introduced. The same gas-substitution procedure was repeated several times before the flask was sealed. After the mixture

was stirred at 60 °C for 24 h, it was hydrolyzed with 10% aqueous solution of HCl at 0 °C. The organic materials were extracted into ethyl acetate. The extract was washed with water and brine, and then dried over anhydrous Na₂SO₄. After evaporation of the solvents, the residue was dissolved in AcOEt (1 mL) and treated with CH₂N₂ in Et₂O, which was prepared from 1-methyl-3-nitro-1-nitrosoguanidine and 40% aq. KOH at 0 °C. The reaction mixture was worked-up according to the standard procedure. The obtained crude material was purified by silica gel column chromatography (hexane/EtOAc=3/1) to afford ester **4aa** as a colorless solid (55.3 mg, 78%). The spectral data of **4aa** obtained by this protocol were fully consistent with those reported in the previous literature.^[5e]

Typical procedure for alkylative carboxylation of allenamides: methyl (2Z)-2-((2-oxooxazolidin-3-yl)methylene)heptanoate (14af) (Table 6, entry 5): The reaction was performed in a 20 mL Schlenk flask with a PTFE J. Young valve (RA-type) equipped with magnetic stirring bar. In a glovebox, the flask was charged with Cu(acac)₂ (33.3 mg, 0.120 mmol), LiCl (35.9 mg, 0.84 mmol), allenamide **11a** (53 mg, 0.42 mmol) in dry DMF (2.0 mL). To this was slowly added BuZnBr (**2f**) (1.0 M in DMA, 1.05 mL, 1.00 mmol). Then the PTFE J. Young valve was closed. The Schlenk flask was took out from the glovebox and the mixture was stirred at 50 °C for 48 hrs. The flask was evacuated and CO₂ (stored in balloon) was quickly introduced. The same gas-substitution procedure was repeated several times before the flask was sealed and the mixture was stirred at 50 °C for 48 hrs. After completion of reaction, the resulting reaction mixture was hydrolyzed with 10% aqueous solution of HCl at 0 °C. The aqueous layer was extracted with ether. The combined organic layers were washed with water and brine. The collected ether extracts were dried over anhydrous Na₂SO₄ and concentrated in vacuo. The residue was dissolved in AcOEt (1 mL) and it was treated with diazomethane in Et₂O. The reaction mixture was worked-up according to the standard procedure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate = 3/1) to afford **12af** (colorless oil, 79.1 mg, 82%). The spectral data of **12af** obtained by this protocol were fully consistent with those reported in the previous literature.^[5f]

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Keywords: ynamides • allenamide • carbon dioxide • carboxylation • copper

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- [22] The present reaction conditions were not ineffective for the reaction of 1,1- and 1,3-disubstituted allenamides with alkylzinc halides possibly due to the steric hindrance.

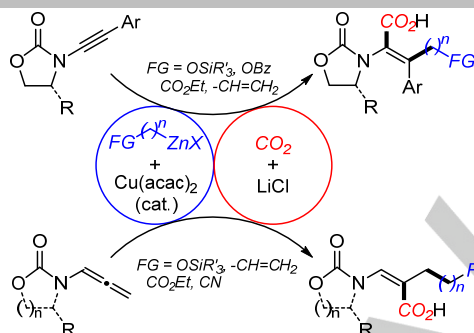
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Double functionalization:

Alkylative carboxylation of ynamides and allenamides with CO₂ and alkylzinc halides via carbozincation of an unsaturated C-C bond and the subsequent carboxylation of the resulting alkenylzinc halides has been developed. This protocol enables carboxylation as well as introduction of functionalized alkyl groups in a highly regio- and stereoselective manner.



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**Alkylative Carboxylation of
Ynamides and Allenamides with
Functionalized Alkylzinc Halides
and Carbon Dioxide by a Copper
Catalyst**

Layout 2:

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